Course notes

The coordinate calculations in this chapter are carried out in an orthonormal frame of space.

Dot product in space

For non-zero vectors $\vec{u}, \vec{v}$, $\vec{u} \cdot \vec{v} = \|\vec{u}\|\|\vec{v}\| \cos\theta$, where $\theta \in [0, \pi]$ is their angle. If one of them is zero, the dot product is 0.

$$\vec{u} \cdot \vec{v} = \tfrac{1}{2}(\|\vec{u} + \vec{v}\|^2 - \|\vec{u}\|^2 - \|\vec{v}\|^2).$$

In an orthonormal frame, for $\vec{u} = (x, y, z)$ and $\vec{v} = (x', y', z')$,

$$\vec{u} \cdot \vec{v} = xx' + yy' + zz'.$$

The dot product is symmetric and bilinear; $\vec{u} \cdot \vec{u} = \|\vec{u}\|^2$. Two vectors are orthogonal if and only if their dot product is zero.

Example. $\vec{u} = (1, 2, 2)$ and $\vec{v} = (2, -1, 0)$ are orthogonal: $2 - 2 + 0 = 0$. Their norms are 3 and $\sqrt{5}$.

Line, plane and normal vector

Two lines are orthogonal if their direction vectors are. A line is orthogonal to a plane if it is orthogonal to two non-collinear directions of that plane. A normal vector to a plane is a non-zero vector orthogonal to all its directions.

Why two directions suffice. If $\vec{n} \cdot \vec{u} = \vec{n} \cdot \vec{v} = 0$, then, for every $\vec{w} = s\vec{u} + t\vec{v}$, $\vec{n} \cdot \vec{w} = s\vec{n} \cdot \vec{u} + t\vec{n} \cdot \vec{v} = 0$.

Vector condition for belonging to a plane

If A belongs to the plane $\mathcal{P}$ with normal vector $\vec{n}$, then $M \in \mathcal{P}$ if and only if $\overrightarrow{AM} \cdot \vec{n} = 0$.

Orthogonal projection and distance

The orthogonal projection H of M onto a plane is the unique point of the plane such that $\overrightarrow{MH}$ is normal to the plane; if M belongs to the plane, $H = M$. The distance from M to the plane is MH. On a line with direction vector $\vec{u}$, the projection satisfies $H \in d$ and $\overrightarrow{MH} \cdot \vec{u} = 0$.

Proof of the minimum. For every point P of the plane, $\overrightarrow{MH} \perp \overrightarrow{HP}$, so $MP^2 = MH^2 + HP^2 \geq MH^2$. Equality holds only for $P = H$. The projection is the closest point of the plane.

Method on a line. If $d = A + \mathbb{R}\vec{u}$, write $H = A + t\vec{u}$ and solve $(A - M + t\vec{u}) \cdot \vec{u} = 0$. Hence $t = \frac{\overrightarrow{AM} \cdot \vec{u}}{\|\vec{u}\|^2}$.

Angles, areas and volumes

For an angle $\widehat{ABC}$, use $\overrightarrow{BA}$ and $\overrightarrow{BC}$: $\cos \widehat{ABC} = \frac{\overrightarrow{BA} \cdot \overrightarrow{BC}}{BA \times BC}$. For the acute angle between two lines, take the absolute value of the quotient. The area of a triangle is $\frac{bh}{2}$; the volume of a tetrahedron is $\frac{\mathcal{A}_{\text{base}} h}{3}$, where h is an orthogonal distance. The units are squared and cubed respectively.

Expansions and loci

By bilinearity,

$$\|\vec{u} + \vec{v}\|^2 = \|\vec{u}\|^2 + 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2, \qquad \|\vec{u} - \vec{v}\|^2 = \|\vec{u}\|^2 - 2\vec{u} \cdot \vec{v} + \|\vec{v}\|^2.$$

We can therefore also calculate $\vec{u} \cdot \vec{v} = \frac{1}{2}(\|\vec{u}\|^2 + \|\vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$. For two distinct points A, B with midpoint I, the locus of points M such that $MA = MB$ is the plane through I with normal vector $\overrightarrow{AB}$: the *perpendicular bisector plane*. Indeed, writing the vectors from I, the expansions give $MA^2 - MB^2 = 2\overrightarrow{IM} \cdot \overrightarrow{AB}$. Equality of the distances is therefore equivalent to $\overrightarrow{IM} \cdot \overrightarrow{AB} = 0$. **Extension, not assessed in 2026–2027.** Two planes are perpendicular when their normal vectors are orthogonal. This property must not be confused with parallelism of planes, which corresponds to collinear normal vectors.